



[The exam text is in English. You should answer the questions in English.]

The exam consists of 13 questions (6 pages). Please first check if any page is missing in the printout you. For the open questions: please keep the handwriting clear and readable.

The exam starts at 9.00 hrs and ends at 12.00 hrs. Participation in the exam requires being present for **at least 1 hour**.

Electronic devices are not allowed and must be stowed.
Points for each open question are indicated.

- Please write readable and write your name and student identification number on every single sheet of paper for unique identification!
- **Don't spend much time writing long stories – that's NOT necessary. Don't panic! You CAN make it!**
- Look at the whole exam first to get an overview, pay attention to the grading scheme of multiple-choice questions, and do those assignments first which you feel are doable for you.
- Good success and all the best for your studies!

Part I - Multiple-choice [50 points].

- Each problem has a variable number of correct statements. Please write your answers in the table given at the end of the question sheet.
- In NO problem, all statements are correct. If you choose all the options, i.e., "ABCD", the score for that question will be ZERO!
- The score for each question will be given according to Morgan method:

$$5 \times \max \{ r / \#correct\ option(s) - w / \#distractors(s), 0 \},$$

where:

- 5 is the total score for each question;
- r is the number of option(s) you select correctly;
- w is the number of option(s) you select incorrectly;
- $\#correct\ option(s)$ is the number of correct options of this question;
- $\#distractors(s)$ is the number of incorrect options/distractors of this question;
- Please DO NOT mark an option if you are not certain with it as a penalty will incur.

Please write the alphabetic characters (e.g., CD) representing your answers of the multiple-choice questions into the table below.

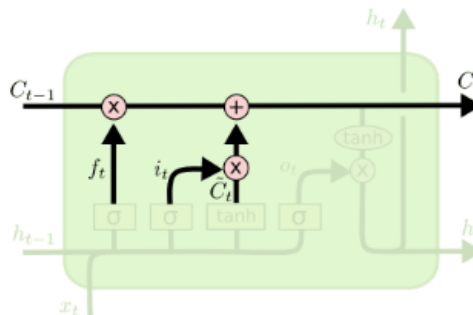
No.	1	2	3	4	5	6	7	8	9	10
Answers	bd	c	c	ac	cd	abc	ab	bcd	bd	ad

1. Regarding the biological and McCulloch-Pitts models, which of the following descriptions are correct? [more than one answer is correct]
 - a. Synaptic strength of the biological neuron is translated into the bias of the McCulloch-Pitt model.
 - b. A step function/threshold unit is used as the activation function in the McCulloch-Pitt model.
 - c. Excitation refers to the negative product of the incoming input and the corresponding synaptic strength.
 - d. Spikes of a biological neuron can be interpreted as firing rates.
2. What is the wrong statement about a Hopfield network? [one answer is correct]
 - a. Hopfield network contains loops in its graphic representation.
 - b. Attractors are local minima of the energy function.
 - c. Hopfield network has multiple layers of neurons.
 - d. The model should be trained using different algorithms than the perceptron model.
3. Which of the following statements is true about Hebbian learning? [one answer is correct]
 - a. Hebb's rule says that the weight between two neurons should be decreased if they are activated simultaneously.
 - b. The learning rate in Hebbian learning is usually larger than one.
 - c. We need apply Hebb's rule on each data point and pass over the entire training set.
 - d. The Hebbian learning process will always stabilize after passing the training set several times.
4. In binary classification tasks using a simple perceptron, which of the following statements are true? [more than one answer is correct]
 - a. The margin is defined as the distance of the closest point to the decision boundary among all correctly classified ones.
 - b. The margin is defined as the distance of the furthest point to the decision boundary among all correctly classified ones.
 - c. Finding the optimal weight vector is translated into a maxmin problem.
 - d. It is often convenient to ask that the product of the predicted and the expected output is smaller than a margin.
5. Which of the following statements are true about multi-layer perceptron? [more than one answer is correct]
 - a. It is well-suited for image classification tasks.
 - b. The XOR problem cannot be solved by multi-layer perceptron.
 - c. Multi-layer perceptron is typically trained by performing gradient descent method.
 - d. The functionality of the hidden layer is to extract features.
6. Which of the following statements are true regarding the backpropagation (BP) algorithm and batch method? [more than one answer is correct]
 - a. The BP algorithm consists of a forward phase and a backward phase.
 - b. Consider a simple perceptron $O = h(w \cdot x + b)$. We can expand the partial derivative $\partial O / \partial w$ as $\partial O / \partial h \cdot \partial h / \partial w$.
 - c. Batch size is the number of data points taken to compute the loss value in each iteration of the BP algorithm.
 - d. An epoch is when half the dataset is passed through the neural network.

7. Which of the following tasks would be more suitable to use RNNs? [more than one answer is correct]
- Forecast the traffic of the next day at a road crossing, given 30-day history thereof.
 - Predict the next word of a sentence based on the previous words.
 - Detect real-world objects in images.
 - Detect anomalous events in bank transactions.
8. Which of the following statements are true about CNNs? [more than one answer is correct]
- In a handwritten digits recognition task (i.e., MNIST), we use the sigmoid unit to obtain the classification probability.
 - Kernels in CNN models are also weights which will be learned via back-propagation process.
 - We can use the cross entropy loss function if the output of the CNN model is classification probability.
 - A tensor is a multi-dimensional array.
9. Which statements are correct about convolution and pooling? [more than one answer is correct]
- Convolution and pooling both contain trainable parameters.
 - Convolution and pooling both extract features from the input image.
 - Convolution and pooling both increases the dimension/size of the input image.
 - Multiple convolutions are performed in each layer of a CNN in order to capture different features of the input.
10. Please calculate the size of the output image (assuming the stride is one) for a given input image and kernel size. Case (1): input size 7x7, kernel size 3x3; case (2): input size 18x18, kernel size 4x4. [more than one answer is correct]
- For case 1, if no padding, the size of the output matrix is 5x5.
 - For case 1, if the padding size is (2, 2), the size of the output volume is 6x6.
 - For case 2, if the padding size is (3, 3), the size of the output volume is 20x20.
 - For case 2, if no padding, the size of the output volume is 15x15.

Part II - Open Questions [50 points]

1. (15 points) Consider the following diagram representing a Long Short-Term Memory (LSTM) unit.



- a) (5) What is the functionality of the variable C_t in the above diagram? Please write it down with your own words.

The variable C_t contains the cell state value. The cell state ensures that gradients don't decay, as it allows for old histories to be forwarded to the current time.

- b) (5) Please write down the expression or operation of the forget gate with the symbols in the above diagram.

$$f(t) = \sigma \left(\mathbf{W}_f [\vec{h}(t-1), \vec{x}(t)] + b_f \right)$$

- c) (5) In contrast to the vanilla Recurrent neural network (RNN), the LSTM relaxes the well-known issue of vanishing/exploding gradients. Please describe in your own words, how LSTM manages to achieve it.

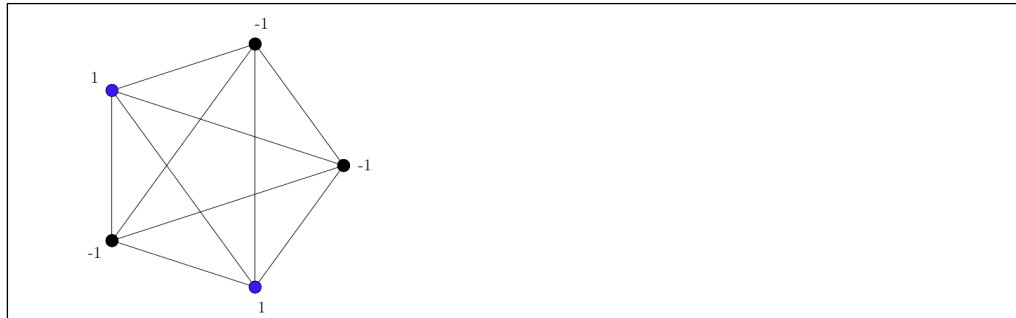
LSTMs leverage gating mechanisms (input gate, forget gate, output gate) to control the flow of information and gradients. They regulate the magnitude of the updates (using element-wise multiplication and addition operations help preserving gradient magnitudes).

The cell state C_t provide a path for gradients to flow more effectively through time, reducing the risk of gradient vanishing.

The gating mechanism also prevent excessive growth of gradients, mitigating the issue of exploding gradients. Useful to this scope is the usage of the tanh activation function, which is bounded.

2. (20 points) Consider a Hopfield network, whose main task is to address the associative memory problem (store a set of p patterns $\vec{\xi}^1, \dots, \vec{\xi}^p$ such that when presented with a new pattern $\vec{x}$, the network responds by producing whichever one of the stored patterns that most closely resembles $\vec{x}$) and perform the following tasks.

- a. (5) Give a graphical representation of a stored pattern $\vec{\xi}^i = [-1, -1, 1, -1, 1]$.



- b. (5) Considering the case of one stored pattern $\vec{\xi}$, what is the condition for the pattern to be a stable pattern, and how should the weights be defined to guarantee that?

The condition for it to be a stable pattern is: $O_i = \text{sgn}(\sum_j w_{ij} \xi_j) = \xi_i$

The weights should be defined as proportional to the node states ($w_{ij} \propto \xi_i \xi_j$), hence $w_{ij} = \alpha \xi_i \xi_j$ and α can be chosen as $1/N$, where N is the number of nodes.

- c. (5) Prove that the definition of the weights given in (b) makes the pattern $\vec{\xi}$ stable.

$$\begin{aligned} O_i &= \text{sgn} \left[\sum_j w_{ij} \xi_j \right] = \\ &= \text{sgn} \left[\sum_j \alpha \xi_i \xi_j \xi_j \right] = \text{sgn} \left[\alpha \xi_i \underbrace{\sum_j 1}_{=N} \right] = \text{sgn} \left(\alpha N \xi_i \right) = \text{sgn} \xi_i = \xi_i \end{aligned}$$

$\Rightarrow O_i = \xi_i \text{ stable!}$

- d. (5) Design a network of 3 neurons (i.e., compute the weights) that remembers a pattern $\vec{\xi} = [1, -1, 1]$. Then prove that $\vec{\xi}$ is stable for the designed network.

$$W = \frac{1}{N} \vec{\xi} \vec{\xi}^T = \frac{1}{3} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & -1 & 1 \end{pmatrix} = \frac{1}{3} \begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$$

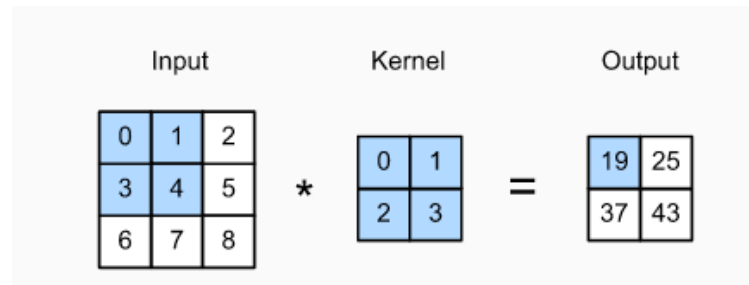
Is $\vec{\xi}$ stable?

$$\begin{aligned} \vec{O} &= \text{sgn}(W \cdot \vec{\xi}) = \text{sgn} \left[\frac{1}{3} \begin{pmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \right] = \text{sgn} \left[\frac{1}{3} \begin{pmatrix} 3 \\ -3 \\ 3 \end{pmatrix} \right] \\ &= \text{sgn} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = \vec{\xi} \end{aligned}$$

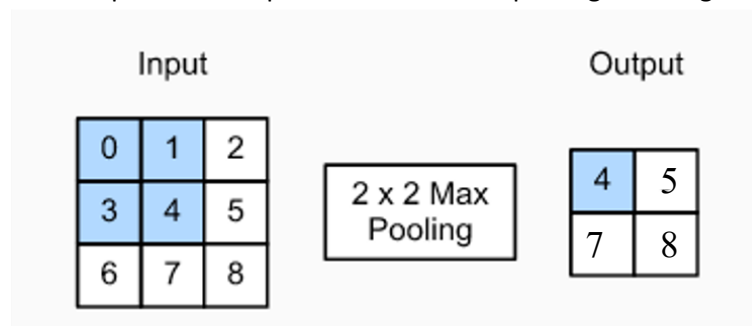
$\Rightarrow$ the answer is YES!

3. **(15 points)** Consider the convolution and pooling operations and answer the following questions:

a. **(5)** Please complete the output of the convolution given in the figure below.

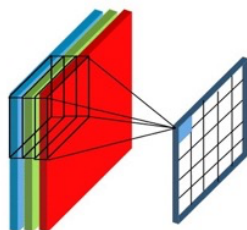


b. **(5)** Please complete the output of the 2 x 2 max-pooling in the figure below.



c. **(5)** Please specify [either graphically or by text description] how to apply the convolution operation to a multi-channel input image, e.g., the red, green, and blue channels in RGB images.

The convolution operation is applied independently to each channel of the input image. For each channel, a different filter/kernel is slid across the input image and the dot product at each position is computed to get the corresponding value in the output feature map.



To summarize:

- Individual filter/kernel (of the same size) for each channel
- Compute (dot product) the resulting value at each pixel for each channel